

Subhan Sajadi

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Education

BEng, Mechanical Engineering (Mechatronics Option), Minor in Computer Science Sep 2022 – Apr 2027
Toronto Metropolitan University • Ontario Scholar • Entrance Scholarship

Coursework: Fundamentals of Robotics, Measurements, Sensors & Instruments, Manufacturing System Control, Vibrations, Microprocessor Systems, Fluid Mechanics, Heat Transfer, Manufacturing Fundamentals

Research Experience

SymBioRobotX Lab, Toronto Metropolitan University Toronto, ON
Undergraduate Research Assistant Jul 2026 – Present

- Built a **MATLAB/Simulink Desktop Real-Time** controller for a CubeMars AK70-9 exosuit actuator with position, velocity, and torque modes and a safe-disable path; real-time kernel mode eliminated a ~ 4 s lag.
- Tuned the actuator's **position PID** (K_p 0.0040 $\rightarrow$ 0.0170), raising closed-loop bandwidth $4.9\times$ (1.07 $\rightarrow$ 5.23 Hz) with no overshoot; now adding an outer Simulink PID loop for torque tracking.
- Helped redesign and assemble the OpenExo-based exosuit's mechanical architecture, from 3D-printed motor cartridges to the sprocket, roller-chain, and Bowden-cable transmission.
- Traced silent driver faults to a $10,000\times$ scaling error and a wrong current-loop ID, bringing torque control online.
- Developed an 8-stage actuator bench test plan with the PI (identification through torque, thermal, and fault tests) and designed a 3D-printed motor testbed with a lever arm for gravity-referenced torque calibration.

Miniaturization in Bioanalysis Lab, iBEST (St. Michael's Hospital) Toronto, ON
Undergraduate Research Assistant May 2026 – Aug 2026

- Developed a two-layer **SU-8 photolithography** process on 100 mm silicon wafers using a MicroWriter ML3 maskless direct-write system to fabricate mother-machine microfluidic devices for single-cell *E. coli* imaging.
- Fabricated PDMS devices end to end (casting, port punching, plasma bonding) and iterated photomask geometry across multiple revisions, writing a **Python** GDS viewer to inspect mask files.
- Diagnosed resist-thickness discrepancies by cross-referencing optical profilometry against spin-speed parameters, and designed discriminating experiments to separate instrument artifacts from coating defects.
- Authored the lab's 30-section cleanroom SOP (v3.0) for the full two-layer workflow with alignment-mark recovery fallbacks, and designed a 3D-printed microscope stage insert in **SolidWorks** for handoff to future students.

Professional Experience

Litens Automotive Group Vaughan, ON
Purchasing Student May 2025 – May 2026

- Built an **Excel/Power Query** cost-benchmarking platform comparing supplier pricing by part type, material, vendor, currency, and historical trend.
- Supported supplier cost forecasting and negotiations contributing to projected multi-million-dollar savings on future automotive programs.
- Evaluated supplier quotes using engineering drawings, BOMs, and cost drivers across machining, forging, casting, and injection moulding; drawing and BOM reviews identified design improvements worth up to 10% in savings.
- Automated purchasing workflows in **VBA**, saving up to 2 hours/day with zero errors on automated POs; managed ~ 200 RFQs and ~ 200 BOM action files and built ERP/PLM training resources.

Celestica Toronto, ON
Commodity Management May 2024 – Aug 2024

- Owned price determination, sourcing, and supplier negotiations for assigned commodities, achieving a 10% increase in cost savings.
- Analyzed BOMs and maintained material price databases to prepare competitive, compliant customer bids, and resolved supplier performance issues through quarterly business reviews.

PALS Learning Centre Toronto, ON
Academic Tutor Sep 2023 – Jun 2025

- Tutored 15+ grades 9–12 students in science and mathematics (~ 6 hours/week), adjusting explanations to each student's level.

Technical Projects & Design Teams

Computer Vision Developer, TMU Wave Autonomous Underwater Vehicle Team

- Developing **OpenCV** detectors for gates, slalom poles, and path markers for an autonomous underwater vehicle, running in a Dockerized ArduPilot SITL and MAVLink simulation stack.

- Proposed a shared detection-output contract (found, center, offset, direction, confidence) and a reusable center-offset utility to simplify integration with the controls and state-machine sub-teams.

1/10-Scale Autonomous Race Car, TMU Velocity (F1Tenth)

- Implemented SLAM-based localization with RViz visualization in **ROS2**, reaching sub-100 ms perception-to-control latency.
- Engineered LiDAR emergency braking with time-to-collision logic, cutting collision events by ~90% in testing.

Smart Rehabilitation Knee Sleeve, True North Biomedical Competition

- Designed and prototyped a smart knee sleeve integrating surface **EMG** and **IMU** sensors to quantify muscle-activation asymmetry and knee range of motion, targeting objective ACL rehabilitation metrics.
- Modelled the full CAD assembly and exploded BOM, with modular sensor mounts, a battery housing, and strain-relieved wire routing for repeated knee flexion.
- Completed a Health Canada/FDA Class I regulatory analysis, specifying electrical isolation and biocompatible materials.

Rocker-Bogie Competition Rover, Take-Off Robotics Competition

- Built a 6-wheel rocker-bogie rover with a servo-driven 3-DOF arm and claw for a three-event competition (obstacle time trials, block placement, laser targeting) on a 5-person team.
- Integrated an adapted rocker-bogie suspension with a custom chassis and arm in **SolidWorks**, and modelled arm workspace, worst-case torque, and tip-over stability in **MATLAB**.
- Programmed **ESP32** PWM motor and servo control with wireless gamepad input; 3D printed and assembled the rover.

Passive Vibration Absorber for a Helicopter Tail Rotor, Vibrations Design Project

- Designed a 2-DOF passive absorber for a 500 g unbalanced rotor mass at 1000–2000 rpm, tuning mass, damping, and frequency ratios with Den Hartog's equal-peak method.
- Reduced peak transmissibility by up to 80% at resonance, verified with **MATLAB** simulations of force transmissibility and amplitude response.

Robotic Gripper Mechanism, Mechanism Design Project

- Designed a 5-bar, 2-DOF gripper driven by dual servos synchronized through gears, with kinematic, static, and torque analysis for a secure grip on objects up to 500 g.
- Simulated motion and workspace in **MATLAB**, modelled the assembly in **SolidWorks**, and 3D printed a functional prototype.

EMG Signal Processing Pipeline, Biosignals Project

- Built a **MATLAB** pipeline (20–450 Hz bandpass, 60 Hz notch, rectification, 100 ms RMS envelope) normalizing tibialis anterior EMG to %MVC.

Leadership

VP Competitions, Ontario Engineering Competition 2027

Jan 2026 – Present

- Co-lead 9 provincial competitions for 300+ Ontario students, directly overseeing 6 competitions and ~20 leads.
- Led the 2026–2027 rulebook and AI-use policy revision, oversaw redesigns of two design challenges after test day exposed flaws, and manage a \$5K build budget.
- Built OEC Tracker to keep ~20 competition leads accountable: every meeting becomes owner-assigned action items, and overdue ones are flagged on Discord, so no follow-up gets lost.
- Prototyped a **FastAPI**/SQLite judging portal with role-based logins, rubric scoring, timers, and CSV export.

VP Events, Mechatronics Course Union

May 2025 – Apr 2026

- Led 4 directors and 6 first-year reps, running 20+ technical events (**SolidWorks**, **Arduino**, **Raspberry Pi**, **ROS2**, **AI/ML**) for 150+ engineering students.
- Built the union's events dashboard (**React/TypeScript**, **Supabase** with row-level security, **Netlify**) with a daily **GitHub Actions** job that ingests union Instagram posts into a cross-union calendar.

VP Competitions, Metropolitan Engineering Competition

May 2023 – Nov 2025

- Ran 9 technical competitions per year across two cycles (18 total, ~16 teams each, ~400 competitors in 2024), leading teams of 11 and then 18 engineers and managing ~\$2K in materials procurement.
- Helped implement judging-scorecard automation (**Google Apps Script**, **Python**) and an inventory website with ~500 users.

Skills

Mechanical: **SolidWorks**, **Fusion 360**, **Onshape**, **AutoCAD**, **ANSYS FEA**, **GD&T**, Engineering Drawings, Metrology, CMM

Electrical & Embedded: **ESP32**, **Arduino**, **CAN Bus (PCAN)**, **I2C/SPI**, **KiCad**, **FPGA/VHDL**

Controls & Analysis: **MATLAB**, **Simulink**, **Simulink Desktop Real-Time**, **Signal Processing**, **Statistical Analysis**

Software: **Python**, **C/C++**, **Java**, **TypeScript/React**, **SQL**, **VBA**, **Bash**, **Git**, **Docker**

Robotics & Vision: **ROS2**, **RViz**, **OpenCV**, **YOLOv8**, **ArduPilot SITL**, **MAVLink**

Fabrication & Lab: **3D Printing (FDM)**, **CNC Machining**, **SU-8 Photolithography**, **PDMS Soft Lithography**, **Cleanroom Protocols**

Business Tools: **Excel/Power Query**, **Power BI**, **SAP/ERP**, **PLM**